

Galois group of x^6+3 over $F = \mathbb{Q}$ or $\mathbb{F}_5$ or $\mathbb{F}_7$. (UCLA QWad, 2022 Fall)
 For each case, determine the splitting field K and evaluate $[K:F]$.
 And, find the Galois group $\text{Gal}(K/F)$.
 (Note: the field of characteristic 0 or finite field are perfect.)

1) $F = \mathbb{Q}$.

By the Eisenstein criterion, all coefficients of x^6+3 except the leading coefficients are contained in $3\mathbb{Z}$, but the square of constant term $3 \notin 3^2\mathbb{Z}$, the given polynomial x^6+3 is irreducible over $\mathbb{Q}$. The x^6+3 has 6 roots,

$$\sqrt[6]{3} \cdot \omega^{2k+1} \quad (k=0,1,\dots,5) \text{ where } \omega = \exp\left(\frac{i\pi}{6}\right) = \frac{\sqrt{3}}{2} + i\frac{1}{2}.$$

Therefore, the splitting field is $K = \mathbb{Q}(\sqrt[6]{3} \cdot \omega, \sqrt[6]{3} \cdot \omega^3, \dots, \sqrt[6]{3} \cdot \omega^5)$,

clearly $\mathbb{Q}(\sqrt[6]{3} \cdot \omega) \subseteq K$, by the definition of generated field.

Meanwhile, $(\sqrt[6]{3} \cdot \omega)^3 = \sqrt{3} \cdot \omega^3 = \sqrt{3} \cdot \exp(i\pi/2) = \sqrt{3} \cdot i$ and $\omega^2 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$ implies $\omega^2 \in \mathbb{Q}(\sqrt[6]{3} \cdot \omega)$, so $K = \mathbb{Q}(\sqrt[6]{3} \cdot \omega)$. Finally, since x^6+3 is irreducible, $[K:\mathbb{Q}] = 6$.

Moreover, $\text{Gal}(K/\mathbb{Q}) \cong S_3$, because: define two automorphisms on K as:

let $\theta = \sqrt[6]{3} \cdot \omega$ and $\lambda = \omega^2$. Then, $\theta^3 = i\sqrt{3}$ and $\lambda = \frac{1}{2} + \frac{\theta^3}{2}$. Now, let

$$\sigma: \theta \mapsto \theta \lambda \quad \text{and} \quad \tau: \theta \mapsto \theta \lambda^2.$$

Then, $\sigma(\lambda) = \sigma\left(\frac{1}{2} + \frac{1}{2}\theta^3\right) = \frac{1}{2} + \frac{1}{2}\theta^3 \lambda^3 = \frac{1}{2} + \frac{1}{2}i\sqrt{3} \cdot \omega^6 = \frac{1}{2}(1 - i\sqrt{3}) = \omega^{10} = \lambda^{-1}$,
 so $\sigma^2(\theta) = \sigma(\theta \lambda) = \theta \lambda \lambda^{-1} = \theta$, it means $\sigma^2 = \text{id}$.

And $\tau(\lambda) = \frac{1}{2} + \frac{1}{2}\theta^3 \lambda^6 = \frac{1}{2} + \frac{1}{2}i\sqrt{3} = \lambda$, so

$$\tau^2(\theta) = \tau(\theta \lambda^2) = \theta \lambda^4, \quad \tau^3(\theta) = \theta \lambda^6 = \theta, \text{ so } \tau^3 = \text{id}.$$

And, $\sigma\tau(\theta) = \sigma(\theta \lambda^2) = \theta \cdot \lambda^{-1} = \theta \cdot \lambda^2$, $(\sigma\tau)^2(\theta) = \sigma\tau(\theta \lambda^2) = \theta \lambda^{-3} = \theta$,
 therefore $\sigma\tau\sigma\tau = \text{id}$, or $\sigma\tau\sigma = \tau^{-1}$. Now, $\text{Gal}(K/F) = \langle \sigma, \tau \rangle \cong D_6 \cong S_3$.

Alternatively, if $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/6\mathbb{Z}$, the abelian group, then every subgroup is normal. Therefore every intermediate field E satisfies E/F is Galois. (In other say, normal)
 Since $\sqrt[6]{3} \cdot \omega^{11}, \sqrt[6]{3} \cdot \omega = \sqrt[3]{3} \in K$, the field $\mathbb{Q}(\sqrt[3]{3})$ is the intermediate field, but as we know, the minimal polynomial of $\sqrt[3]{3}$ is x^3-3 over $\mathbb{Q}$, but $\mathbb{Q}(\sqrt[3]{3})$ does not have other roots of x^3-3 . So, it is contradiction.

By the knowledge in the group theory, there are only two group of order 6, S_3 and $\mathbb{Z}/6\mathbb{Z}$, so $\text{Gal}(K/\mathbb{Q}) \cong S_3$. □